Reliability-Aware Fusion of Token Evidence for Image-Text Verification in Unified Discrete Diffusion

Shlok Goenka, Subramaniyaswamy Vairavasundaram, and Balaji G N

***Abstract*—Image-text verification, choosing the true caption of an image from two candidates, becomes harder when the image is damaged. We study it in one frozen unified discrete diffusion model, UniDisc, which scores every image token given a caption. Each token's log-probability ratio between the true and a wrong caption is evidence, and the standard rule sums it with equal weights. We learn a per-token reliability weighting of this evidence from caption-free features, fit it on design images that include the noise level later tested, and freeze it. A preregistered one-sided exact paired test on fresh Common Objects in Context validation images under Gaussian noise at one level confirmed that the frozen weighting raised verification accuracy, also with wrong captions matched in length. On clean images it was non-inferior within a preregistered margin, fixed after an exploratory clean-image interval had been seen. In exploratory comparisons, a pixel denoiser before scoring did better under Gaussian noise, and combined with the weighting it did worse under Joint Photographic Experts Group compression. A linear logit on the same features did about as well under noise and better under compression. A sweep on the same images found the gain at milder and stronger noise, but at the tested level it did not clearly exceed the clean-image gain. The evidence covers one model, one dataset, and one tested noise level; it does not show that the weights track token reliability (the name describes the design) or that the gain transfers to other models or corruptions.**

***Index Terms*—Denoising diffusion, discrete diffusion, information fusion, masked diffusion, multimodal learning.**

I. Introduction

A unified masked diffusion model places text tokens and image tokens in one sequence and is trained to predict any masked token from any visible subset, the mask token standing for "unknown" [1]–[3]. Such models generate and understand both modalities with one network [4]–[6]. With the image tokens masked and a caption visible, the same network assigns a probability to each observed image code, with no external encoder involved. In this paper, *image-text verification* means this image-direction decision: for one image and two candidate captions, the true caption is chosen if the observed image tokens are, summed over tokens, more probable under it than under a wrong caption (Section III).

Generative models are already used as scorers in this way. Diffusion classifiers compare conditional denoising errors across labels [7], [8], image-text matching has been scored by a conditional error normalised by the unconditional one [9], and the authors of UniDisc give examples in which damaged image-caption pairs are repaired by masking, re-infilling, and selection by the model's own likelihood [4]. This paper studies evidence that the unified model itself computes, so its behaviour is a property of the scorer, not of an attached encoder. Images can be damaged before a model sees them, by compression, sensor noise, or editing; corruption benchmarks treat such damage as a test condition [10], and multimodal fusion methods are evaluated under it [11].

Problem: The per-token evidence is a sum of terms of unequal quality. A damaged image changes some image tokens and leaves others intact, and the model's evidence at a changed token need not point toward the true caption. When evidence comes from sources of unequal quality, the classical combination rule gives each source a weight that reflects its reliability [12], [13]; here each image token is such a source, and the unweighted sum gives all of them the same weight. The problem studied here is whether a per-token weighting of the likelihood evidence of a frozen unified masked diffusion model, fitted in advance and then fixed, raises image-text verification accuracy on fresh images when the image is damaged before tokenisation (Fig. 1). The question was preregistered with its own plan, test images, and decision rule (Section III-E): does one frozen, learned per-token weighting of the evidence raise image-direction verification accuracy on fresh images under Gaussian noise?

Contributions:

1) *A measurement of the evidence.* We define the per-token image-direction evidence Δ_j of a unified masked diffusion model and the unweighted sum that decides between two captions, with no training or finetuning of the model and no attached encoder (Section III).
2) *A learned reliability weighting, frozen before any test image was chosen.* Learned reliability-aware fusion (LRF), a logistic generalised additive model, maps 35 caption-withheld features of each image token to a non-negative weight on its evidence. It was fitted on 500 design images, frozen, and pinned by SHA-256 before the test images were selected (Section IV).
3) *A preregistered confirmation.* On a fresh preregistered fold, the frozen LRF raised image-direction verification accuracy under Gaussian noise ($\sigma = 0.1$) from 0.733 without reliability weighting to 0.802 (exact one-sided McNemar $p = 0.0005$), including with length-matched negatives; on clean images it lost no more than three percentage points (0.03 in accuracy; 90% interval of the net change 0.009 to 0.053; margin fixed after an exploratory clean interval had been seen). This holds in one pretrained unified masked diffusion model (UniDisc,

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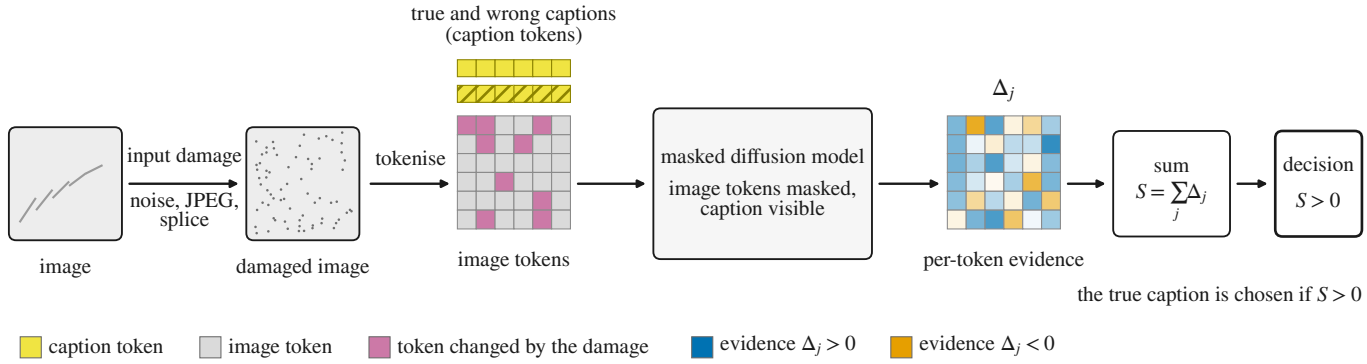


Fig. 1. Image-text verification by per-token evidence in a masked diffusion model (conceptual drawing with a fixed token pattern; no data). A possibly damaged image is tokenised; with the image tokens masked and each caption visible, the model gives the evidence Δ_j, the log-probability ratio of the observed image token under the true and a wrong caption, and the unweighted sum $S = \sum_j \Delta_j$ chooses the true caption if $S > 0$.

1.4B parameters), on 450 COCO val2017 images not used to design or fit the method, with Gaussian noise at one severity.

4) *A preregistration record.* The plans, hash pins, every deviation, and the full outputs of every preregistered test are reported (Section V-C; supplement S-I and S-II), together with the cost of the method and, for reference only, the accuracy of two image-text encoders on the same items (Section IV).

Scope of this paper: This paper reports one preregistered measurement on one model and does not run a race between verifiers. Its object is the evidence that a unified masked diffusion model computes about its own inputs, and its question is whether a rule fixed in advance changes the decision, by an exact paired test, on images the rule never saw. The paper does not claim a verifier that ranks with dedicated encoders, or any result for another model, dataset, severity, or kind of negative caption; it does not claim that the learned weights track reliability or corruption.

II. Related Work

Generative and likelihood scoring: Diffusion models are trained by maximising a variational lower bound on the data log-likelihood, as surveyed for continuous diffusion in image super-resolution [14], so a trained model can be read as a scorer of its own inputs. Text-to-image diffusion models have been used as zero-shot classifiers whose weights over diffusion timesteps are learned from held-out labelled data [8], including with few labels [15]. For image-text matching, the conditional error has been normalised by the unconditional one [9], and VQAScore scores alignment by the probability of "Yes" that an image-to-text model assigns to a question about the pair [16]. In text, masked language models score a token with the token masked [17], [18]. The evidence used in this paper belongs to this family: a per-token log-likelihood ratio of the observed image codes under two captions, computed by the masked denoiser of a unified model.

Masked discrete diffusion and unified models: Absorbing-state discrete diffusion [1], [19] was simplified into masked diffusion language models [2] and scaled [20]; the denoiser of such a model represents the conditional distribution of a clean token given the unmasked context [3]. MaskGIT generates image tokens by masked prediction [21], and unified text-image models place several modalities in one sequence [4]–[6], as does Muddit, published as "Beyond text-to-image: Liberating generation with a unified discrete diffusion model" [22]. LaViDa adds a vision encoder to a diffusion language model [23], and Lumina-DiMOO is a further omni diffusion language model for multimodal generation and understanding [24]. In this journal, discrete-diffusion image captioning has been studied in a setting where noise in the diffusion process weakens sentence information, and a feature-mask and gated-fusion design improves image-text alignment [25]. Many methods remask tokens that the sampler itself produced, by confidence, learned quality scores, or critics, including cross-modal confidence in multimodal masked diffusion models [26]–[32]. Work on damaged *input* tokens is scarcer: Trust but Verify detects corrupted context tokens in a finetuned text-only masked diffusion model and refills them [33], and dropout decoding masks visual tokens that are uncertain under a logit-lens projection [34]. None of them reports a verification decision under image corruption in a unified masked diffusion model.

Reliability-aware and dynamic fusion: When evidence comes from sources of unequal quality, the classical combination rule weights each source by its reliability [12], and classifier combination studies the rules for doing so [13]. In multimodal learning the weights change per input: dynamic, uncertainty-driven weighting of modalities has been analysed for low-quality data, with theory on when it is preferable to static weighting [35]. A recent survey organises fusion on low-quality data into noisy, incomplete, imbalanced, and quality-varying settings [36]. Trusted multi-view classification fuses per-view evidence dynamically to handle unreliable views [37], and evidential methods attach an estimated importance or reliability to each modality before combining [38]–[40]. Separate data and model uncertainties can also set each modality's weight [41], whose authors leave finer-grained uncertainty modelling to future work. When modalities are missing, recovered modalities can be selected at inference by a loss-derived reward on whole

modalities [42]; our weights act on single tokens inside one model. Papers in this journal take the same view: a per-modality confidence score (an energy score) lets the fusion rely less on low-quality modalities [43]; evidence theory quantifies the uncertainty of each prediction and fuses evidence across propagation depths [44]; evidential deep learning suppresses unreliable modality-wise pseudo-labels in audio-visual video parsing [45]; and counterfactual samples in a shared multimodal latent space identify the input features that contribute to high predictive uncertainty [46]. At a finer grain, attention-based multiple-instance pooling learns instance weights from instance features under a bag-level loss [47], and a token weight has been derived from a model's logits with and without the image, as a training-time reweighting [48]. The closest precedents are thus learned timestep weights [8], [15], instance weights under a bag-level loss [47], and reliability weights of modalities or views [35], [37]. Token-level confidences of a captioning vision-language model, aggregated algebraically or by a learned estimator, have also been used to judge whether a caption is correct for its image [49]; that work scores the caption's tokens given the image and does not study image corruption. To our knowledge, no earlier work learns per-token, caption-withheld weights for the log-likelihood-ratio evidence of a unified masked diffusion model and tests them on fresh images under image corruption. This statement is limited to a keyword search of arXiv and of title-level records of IEEE-published journals and proceedings, logged in October 2026; the search did not cover full texts, the pages of IEEE Xplore itself, or the submission lists of open-review venues.

Conformal risk control: Conformal prediction gives distribution-free guarantees under exchangeability [50], [51]; conformal p-values give false-discovery control for outlier detection [52], and conformal risk control bounds the expected value of a monotone loss [53]. Conformal loss-controlling prediction, published in this journal, extends conformal prediction from miscoverage to a general loss for each test object, with a finite-sample guarantee under exchangeability [54]. Conformal risk control has been applied, with masking-derived CLIPScore distributions and a threshold set on foil-labelled data, to flag erroneous caption words given an image [55]. The weighting studied here carries no such guarantee, and its evidence is a preregistered paired test.

Corruption benchmarks and preregistration: Common-corruption benchmarks test models under non-adversarial image corruptions [10], and a corruption benchmark for object detection released an implementation of such corruptions at graded severities [56], which this paper uses for its other noise types. For image-text models, 17 such corruptions at five severities have been benchmarked on COCO retrieval, without a unified masked diffusion model [57]. Preregistration records the questions and the analysis plan before the outcomes are observed, which separates prediction from postdiction [58]; the form used here is defined in Section III-E.

III. Setting

A. Model and Tokens

The model is UniDisc [4], a unified masked discrete diffusion model with 1.4B parameters, used as released with no training or finetuning here. A pair is a caption of ℓ text tokens and an image of m image tokens in one sequence. Images are tokenised at 512 px by the VQ-16 tokeniser of LlamaGen [59] into a 32×32 grid, so $m = 1\,024$ tokens, each one of 16 384 codes; captions have 5 to 32 text tokens. The network p_θ maps a partly masked sequence to a categorical distribution at every masked position. The backbone runs in `bf16`, and every log-probability is a 32-bit log-softmax of the logits restricted to the vocabulary of the scored modality. UniDisc's denoiser has no time input in the released configuration: time conditioning is off, the official inference configuration passes a null noise level, the backbone builds no timestep embedder without time conditioning, and the released checkpoint contains no time-embedding weights. The scorer uses the authors' null-sigma inference path. This is a property of the released configuration and checkpoint, read in the authors' code at a pinned commit, and not a statement about UniDisc in general (evidence and limits in supplement S-I).

B. Per-Token Evidence and the Unweighted Decision

Write $\tilde{v} = (\tilde{v}_1, \ldots, \tilde{v}_m)$ for the observed, possibly damaged image tokens, c^+ for the true caption, and c^- for a wrong caption. For a caption c, let $\Lambda_j^c[y]$ be the log-probability of code y at image position j in one pass of p_θ with every image token masked and c visible. The *image-direction evidence* of token j is

$$\Delta_j = \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j], \tag{1}$$

the log-likelihood ratio of the observed code under the two captions; $\Delta_j > 0$ when the observed code is more probable under the true caption. Because every image token is masked in these passes, their logits do not depend on the image, which enters only through the codes at which the log-probabilities are read. The verification statistic is the unweighted sum, and the decision chooses the true caption when the sum is positive:

$$S = \sum_{j=1}^{m} \Delta_j, \qquad Y = \mathbb{1}\big[S > 0\big], \tag{2}$$

with a tie counted as wrong. The sum of one-pass conditionals is a factorised surrogate for the joint log-likelihood ratio; an exact ordered likelihood was not computed. Image-text verification accuracy is the mean of Y over the items of a test set, and a net accuracy change is the accuracy of the rule under test minus that of (2) on the same items, so a positive value favours the rule under test. The caption-visible passes are the only inputs of Δ_j. Two scores are computed with the caption masked: the position marginal $D_j = \log p_\theta(\tilde{v}_j \mid \text{all tokens masked})$, and B_j, the log-probability of $\tilde{v}_j$ when the tokens of its group in a random partition of the image positions are masked and the rest of the image is visible. (1) and (2) are the whole measurement of the unweighted rule; Fig. 1 draws it.

C. Corruptions

Three corruptions act on the pixels before tokenisation. They are JPEG compression at quality 10, Gaussian noise with standard deviation $\sigma = 0.1$ on pixels scaled to $[0, 1]$ (clipped and rounded to 8 bits), and a 128×128-pixel splice, in which

a region of a donor COCO image is copied to a uniformly random location. The noise draw, the donor image, and the splice location derive from the item, so each image has one corruption realisation.

D. Data and Fresh Images

Images and captions come from the COCO val2017 split [60]. An item is one image with its true caption and one wrong caption, the caption of another image; Section IV-C states how the wrong caption is drawn. Test images are fresh in a defined sense: they are disjoint, by COCO image id, from every image that was used to fit or design the rule. The test fold also excludes every image used anywhere in the project, including the earlier preregistered stage that supplied the design images (Section IV-A), and every SugarCrepe image [61] that the project had prepared.

E. Tests, Multiplicity, and Preregistration

Every test compares two decision rules on the same n items. Let b be the number of items that the rule under test decides correctly and the reference rule wrongly, and c the reverse. The net accuracy change is $(b-c)/n$, and the exact McNemar test [62] conditions on the $n_d = b + c$ discordant items:

$$p_{\text{one}} = \Pr\left[\text{Bin}(n_d, \tfrac{1}{2}) \geq b\right],$$
$$p_{\text{two}} = \min\left\{1,\ 2\min\left(p_{\text{one}}, \Pr[\text{Bin}(n_d, \tfrac{1}{2}) \leq b]\right)\right\}. \quad (3)$$

The one-sided form of (3) is used only for hypotheses with a preregistered direction, and every other test is two-sided. Within a family of hypotheses, the familywise error rate is held at 0.05 by Holm's step-down procedure [63]. Intervals for the net accuracy change are paired percentile bootstrap intervals over items. This paper contains four preregistered families (L, L-NI, L-LEN, and LG; Section IV-C), each tested at its own level with no adjustment across families; under a global null with independent families, the chance of at least one false rejection is $1 - 0.95^4 = 0.19$. Counting also the families of the earlier study (17 in all), a Bonferroni factor of 17 on the primary one-sided p still gives 0.0086. Supplement S-I lists every family with its test and its wording limits, and the Results give the outcomes.

Preregistered has one meaning in this paper: a dated analysis-plan addendum was committed to a version-controlled repository before the first test item was scored, with the tested object pinned by hash, and the order of the commits is the evidence; no third-party timestamp exists. The repository history was first pushed to a code-hosting service on 30 October 2026, after the preregistered tests reported here had been run; its push log is the only third-party record, and it dates the push, not the commits. The plans, pins, and their amendments are listed in supplement S-I. The commit order, the hash checks, and the data checks were verified by automated audits run in the authors' own pipeline (not independent reviews).

IV. Learned Reliability-Aware Fusion of Token Evidence

This section tests one weighting of the evidence Δ_j (Section I); it is called learned reliability-aware fusion (LRF), a name that describes its design, not a measured property.

A. Provenance of the Tested Method

The 500 design images came from an earlier preregistered study of the same model that tested a different question; its preregistered primary result was negative (no improvement was detected), and that study is not reported in this paper. The method was designed after that study had been analysed, on those images, so the design was not preregistered. Before the design, the analysts had seen the results of that study's analyses, which are not reported here, and an exploratory learned-weighting analysis on the design images. In that analysis four learned-weighting families (a logistic generalised additive model, gradient-boosted trees, a multilayer perceptron, and a grid model) were compared, and the logistic generalised additive model was selected by inner cross-validation. Two other method designs and two other project routes were considered and not pursued. A first freeze was found to read the true caption's length through the layout of its inputs; the caption slot was therefore fixed, and the method was re-scored and re-frozen at that slot length, a choice declared before any re-frozen result existed. Before the re-freeze the analysts had also seen further outputs of that earlier study; the re-freeze protocol itself had been fixed earlier and was not changed. The configuration was frozen (SHA-256 `db7fdb4e...`, commit `24d36a8a`) before any image of the fresh fold was selected. The test was preregistered (a dated addendum to the analysis plan, commit `941a260b`) with a rule that no further confirmatory study would be added. All LRF numbers on the design images are post hoc and exploratory and appear only in supplement S-II.

B. The Frozen Weighting

Reliability-weighted fusion gives each source of evidence a weight that depends on its reliability [12], [13], [35], [37], [43], and attention-based multiple-instance pooling learns instance weights from instance features under a bag-level loss [47]. LRF applies this construction to the per-token evidence of a unified masked diffusion model: an additive model maps features of each image token to a non-negative weight, and the decision is the sign of the weighted evidence.

Features: Each token has 35 features in the seven groups of Fig. 2 (formulas in supplement S-II). One group holds seven token scores (five computed by the model with the caption masked, a patch-texture score, and a codebook-neighbour distance), one holds the token's grid position, and the other five summarise the scores over a neighbourhood or the image. No feature uses Δ_j, the caption-visible log-probabilities, the captions, the corruption labels, or the identity of the condition. Features are standardised with the means and standard deviations of the design tokens.

Weight model and decision: For token j with standardised features $x_{j1}, \ldots, x_{j,35}$, the weight is a logistic generalised additive model [64] with a hinge basis,

$$w_j = \text{sigm}\left(\beta_0 + \sum_{k=1}^{35} a_k x_{jk} + \sum_{k=1}^{35}\sum_{l=1}^{Q} c_{kl}\,(x_{jk} - t_{kl})_+\right), \quad (4)$$

Algorithm 1: Decision rule of the frozen LRF

Input : observed image tokens $\tilde{v}_j$, $j = 1, \dots, m$ ($m = 1\,024$), and the caption pair (c^+, c^-)

Output : the chosen caption: c^+ if $M > 0$ (then $Y_{\mathrm{LRF}} = 1$), otherwise c^- (a tie counts as wrong), and the margin M, (5)

Data : frozen μ_k, s_k (design-token mean and standard deviation of feature k), the $Q = 8$ knots t_{kl} of each feature as stored in the frozen model (`spec.json`), and β_0, a_k, c_{kl} (SHA-256 `db7fdb4e...`)

compute the seven token scores (caption slot filled by 13 padding tokens) and the item-level aggregates (windows, ranks, image means; supplement S-II);

for $j \leftarrow 1$ **to** m **do**

  $\Delta_j \leftarrow$ evidence of token j (Section III-B), each caption at its own length;

  $x_{jk} \leftarrow$ (feature k of token $j - \mu_k)/s_k$ for $k = 1, \dots, 35$, from the scores, aggregates, and grid position;

  $w_j \leftarrow \mathrm{sigm}\left(\beta_0 + \sum_k a_k x_{jk} + \sum_{k,l} c_{kl}(x_{jk} - t_{kl})_+\right)$ (4);

$M \leftarrow \sum_j w_j \Delta_j / \sum_j w_j$;

choose c^+ if $\sum_j w_j \Delta_j > 0$ and c^- otherwise;

where sigm is the logistic function and the knots t_{kl} are the $l/(Q+1)$ quantiles of feature k over the design tokens. The decision and the normalised margin are

$$Y_{\mathrm{LRF}} = \mathbb{1}\Big[\sum_{j=1}^{m} w_j \Delta_j > 0\Big], \qquad M = \frac{\sum_j w_j \Delta_j}{\sum_j w_j}, \tag{5}$$

with a tie counted as wrong; Algorithm 1 lists the steps of the frozen rule.

Objective and tuning: The parameters minimise, over the item-conditions of the design set,

$$\frac{1}{N}\sum_{i=1}^{N} \log\left(1 + e^{-M_i/T}\right) + \lambda\left(\|a\|^2 + \|c\|^2\right), \tag{6}$$

a softplus loss on the margin M_i of (5), which is positive when the true caption is chosen. The temperature is $T = 0.170$; optimisation is AdamW without weight decay on mini-batches of 32 item-conditions. The configuration selected by successive halving on three inner folds of the 500 design items (search in supplement S-II; $Q = 8$ knots, $\lambda = 0.0027$, learning rate 0.0169) was refitted on all 500 items for 40 epochs with a fixed seed. The training data were the 500 design images, each scored clean, after JPEG compression at quality 10, with Gaussian noise of standard deviation 0.1, and with a 128-pixel splice; these are also the four tested conditions.

Fixed caption slot and hashes: The caption-masked scores are computed with the caption slot filled by 13 padding tokens, so they do not depend on the length of either caption; Δ_j keeps each caption at its own length (invariance checks in supplement S-II). The model file and every input of the scorer are pinned by SHA-256 and checked before each item (supplement S-I). Fig. 2 draws the model; the learned shapes of the six features with the largest effect range are in supplement S-II, and Fig. 4 shows three fresh-fold items.

C. Preregistered Protocol

Fresh fold and sample size: The fold was built after the object commit: 533 COCO val2017 images in a seeded order, drawn from the 2 449 eligible images that remain after the exclusions of Section III-D (2 526 image ids, some outside the captioned set) and the removal of images without instance annotations. Each image has one seeded true caption of 5 to 32 tokens, a wrong caption drawn by the preregistered derangement rule (a random caption of another fold image sharing no annotated COCO category with it), and a second, length-matched wrong caption drawn under the same category rule. The fold, its eight-check leak test (with an injected leak as positive control), and the code passed automated audits run in the authors' own pipeline (not independent reviews) before the first item. A preregistered power rule, committed before the first item, fixed the test size as a prefix of the ordered list, so it could not select items: $n_L = 450$ (its inputs are in supplement S-II).

Arms, families, and tests: The arms are none (weight one on all 1 024 tokens) and LRF, under clean images and the three corruptions of Section III-C. Family L has three hypotheses, tested in a fixed sequence followed by Holm (familywise level 0.05). The primary L1 tests LRF against none under noise by the exact one-sided McNemar test at 0.05 and is "confirmed" if and only if $p \le 0.05$ and $b > c$. Only if L1 is confirmed are L3 (JPEG) and L4 (splice) tested, by exact two-sided McNemar tests with Holm over two. A gain claim in the title, abstract, or highlights required L1 one-sided $p \le 0.025$. L-NI (one hypothesis, regardless of L1) tests non-inferiority on clean images with margin -0.03, read from the lower bound of the 90% paired bootstrap interval; the margin was fixed after an exploratory clean interval had been seen. L-LEN (one hypothesis) repeats L1 with the length-matched wrong captions; items whose length-matched caption still differs in length are excluded and counted. Family LG (Holm over three, one-sided 0.05) tests LRF against none on other noise types (shot, impulse, speckle; imagecorruptions severity 3) [56] on the first 300 fold items. Shot and speckle noise are close relatives of Gaussian noise, impulse noise is the structurally different cell, and severity 3 is not matched to $\sigma = 0.1$. According to the operations log, no output was read before its run was complete.

D. Results

Fig. 3 shows the net change and its interval for each family-L hypothesis and Table I gives every preregistered test; each family is at its own level, with no adjustment across families (Section III-E). The remaining quantities that the plan reports with every test (n_d, 90% intervals, the Clopper–Pearson interval of b/n_d and the mid-p values) are in supplement S-II. On a fresh preregistered fold, the frozen LRF raised image-direction verification accuracy under Gaussian noise ($\sigma = 0.1$) from 0.733 without reliability weighting to 0.802 (net 0.069, 95% interval 0.029 to 0.109; exact one-sided McNemar $p = 0.0005$; preregistered one-sided $\alpha = 0.05$), including with length-matched negatives (0.708 to 0.784, net 0.076, 0.033 to 0.120; one-sided $p = 0.0004$; $n = 449$, one item excluded for a length mismatch). L1 is therefore confirmed. This result holds in one pretrained unified masked diffusion model (UniDisc, 1.4B parameters), on 450 COCO val2017 images not used to design or fit the method, with Gaussian noise at one severity.

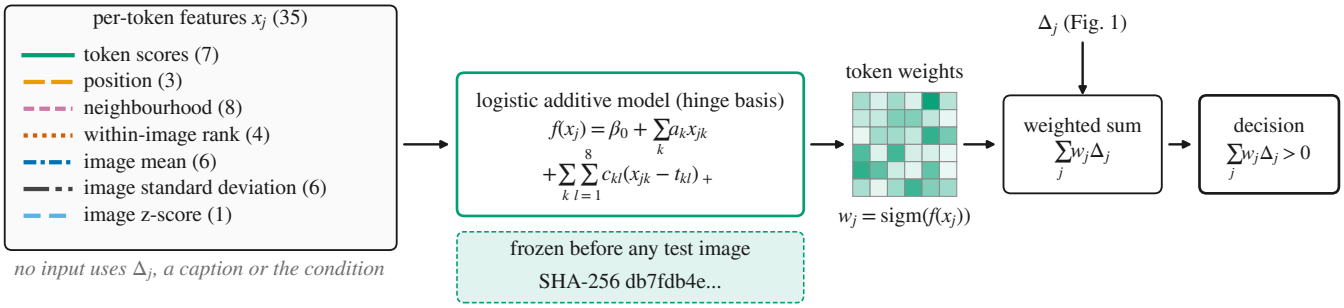


Fig. 2. Learned reliability-aware fusion (LRF) and its frozen weight model (schematic; no data; fixed weight grid). Each image token has 35 features x_j in seven groups (counts as printed; formulas in supplement S-II); no feature uses Δ_j, a caption, or the identity of the condition. A logistic additive model with a hinge basis (4) gives the weight w_j, and the true caption is chosen if $\sum_j w_j \Delta_j > 0$ (5).

TABLE I
PREREGISTERED FAMILIES ON THE FRESH FOLD: ACCURACY WITHOUT AND WITH LRF, DISCORDANT COUNTS, p-VALUES, AND DECISIONS

Hypothesis	n	Acc. none	Acc. LRF	b	c	Net (95% interval)	p one-sided	p Holm	p two-sided	Preregistered decision
L1: noise, $\sigma = 0.1$	450	0.733	0.802	58	27	**0.069** [0.029, 0.109]	0.0005	–	0.001	**confirmed (one-sided)**
L3: JPEG, quality 10	450	0.849	0.864	27	20	0.016 [−0.013, 0.044]	0.191	0.382	0.382	not rejected
L4: splice, 128 px	450	0.896	0.927	24	10	**0.031** [0.007, 0.058]	0.012	0.049	0.024	**rejected (Holm)**
L-NI: clean images	450	0.887	0.918	27	13	0.031 [0.009, 0.053]†	–	–	–	non-inferior (lower bound 0.009 above −0.03)
L-LEN: noise, length-matched negatives	449	0.708	0.784	67	33	**0.076** [0.033, 0.120]	0.0004	–	0.0009	**rejected (one-sided)**
LG: shot noise	300	0.673	0.767	43	15	**0.093** [0.047, 0.143]	0.0002	0.0005	0.0003	**rejected (Holm, one-sided)**
LG: impulse noise	300	0.700	0.750	35	20	**0.050** [0.000, 0.100]	0.029	0.029	0.058	**rejected (Holm, one-sided)**
LG: speckle noise	300	0.697	0.757	32	14	**0.060** [0.017, 0.103]	0.006	0.011	0.011	**rejected (Holm, one-sided)**

b: items LRF answers correctly and the unweighted rule incorrectly; c: the reverse; net $= (b - c)/n$. Intervals: 95% paired percentile bootstrap (10 000 resamples), except † (L-NI): 90%, margin −0.03 fixed after an exploratory clean interval had been seen; L-NI's test is the lower 90% bound against the margin, so no p is printed. p: exact McNemar; p Holm: L3 and L4 after L1 (two-sided), LG over its three cells (one-sided). Bold: null rejected at the preregistered level.

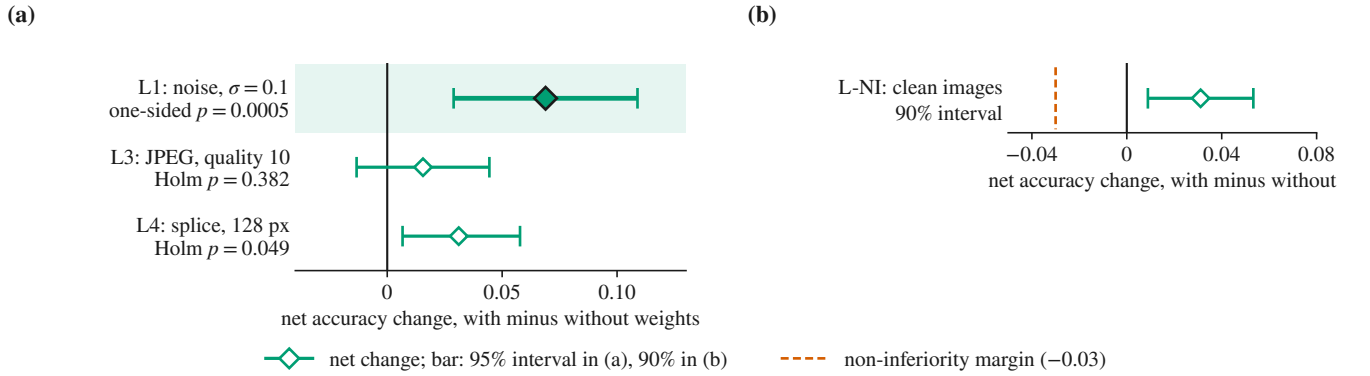


Fig. 3. Family L and L-NI on the fresh fold: net accuracy change of the frozen LRF, with minus without reliability weights, $n = 450$. (a) Diamond: net change; bar: 95% paired percentile bootstrap interval. L1 (Gaussian noise, standard deviation 0.1; primary, shaded): exact one-sided McNemar p; L3 (JPEG quality 10) and L4 (128-px splice): Holm-adjusted two-sided p. (b) L-NI (clean images; own family): net change, 90% interval, and the margin −0.03 (dashed), fixed after an exploratory clean interval had been seen.

On clean images LRF lost no more than three percentage points (0.03 in accuracy; 90% interval of the net change 0.009 to 0.053); the margin was fixed after an exploratory clean interval had been seen. The clean-image estimate, descriptive, was positive (0.887 without weighting, 0.918 with; net 0.031); superiority on clean images was not a preregistered hypothesis. Supplement S-II plots the discordant items of L1 and of the clean cell by margin (descriptive). Fig. 4 shows three items under noise, one of each of three kinds, chosen by a rule fixed before the figure was made, after the confirmatory result: both rules correct, only LRF correct, and only the unweighted rule correct. Items of these three kinds were 303, 58, and 27 of the 450, and both rules were wrong on the other 62, so the rows are examples, not a sample. The maps are descriptive and do not show that the weights track reliability or corruption.

Under JPEG quality 10 no difference was detected (net 0.016, 95% interval −0.013 to 0.044; Holm-adjusted $p = 0.382$). Under the 128-pixel splice LRF raised accuracy from 0.896 to 0.927 (net 0.031, 0.007 to 0.058); the Holm-adjusted p is 0.049, 0.001 below the 0.05 level (unadjusted two-sided $p = 0.024$). No tested condition showed a reduction: the lowest 95% bound was −0.013, under JPEG.

None of the three other noise types (shot, impulse, speckle; imagecorruptions severity 3) was a training condition of the frozen weighting (training conditions: clean, JPEG quality 10, Gaussian noise 0.1, 128-pixel splice). Shot and speckle noise

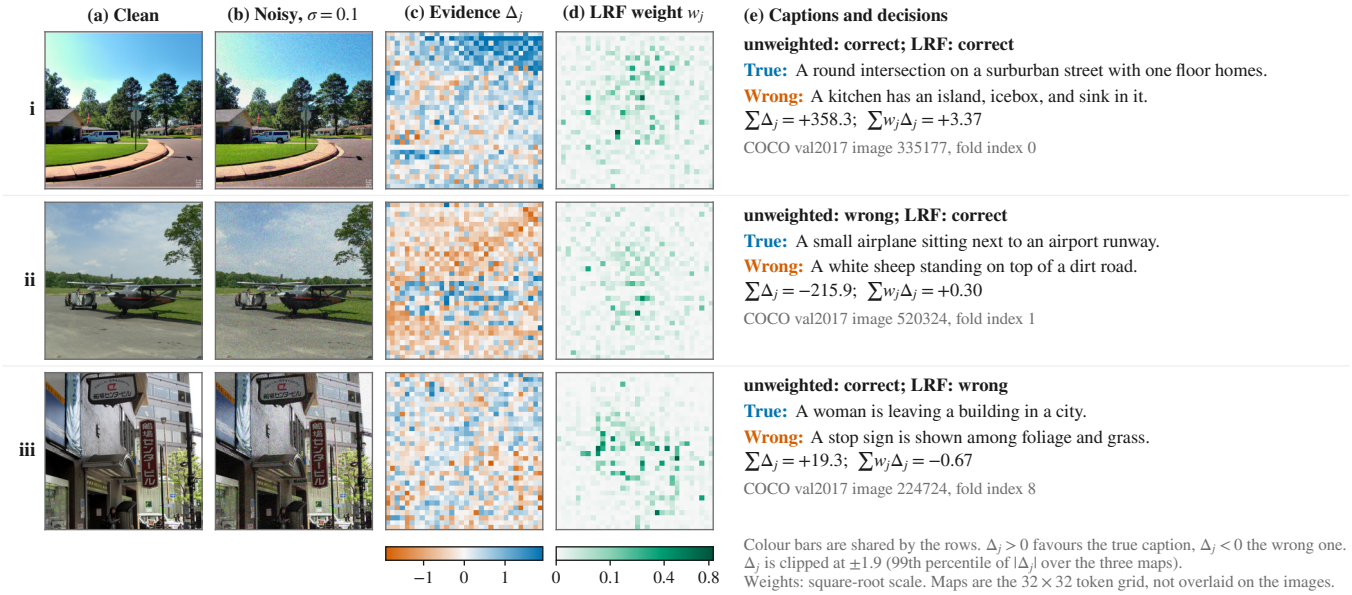


Fig. 4. Three fresh-fold items under Gaussian noise ($\sigma = 0.1$), one of each kind, chosen by a rule fixed before the figure was made, after the confirmatory result: the first item in fold-file order for which the unweighted rule and LRF are both correct (i), only LRF is correct (ii), and only the unweighted rule is correct (iii); the three kinds occur 303, 58, and 27 times among the 450 items (both rules are wrong on 62). Columns: clean image, noisy image, per-token evidence Δ_j and LRF weight w_j on the 32×32 token grid (colour scales in supplement S-VI), and the true and the wrong caption with both decisions and sums. Descriptive; not evidence that the weights track reliability or corruption. Images and captions are from COCO val2017 [60] (identifiers 335177, 520324, 224724; licences in supplement S-VI); noisy panels modified.

are close relatives of Gaussian noise, impulse noise is the structurally different cell, and severity 3 is not matched to $\sigma = 0.1$; no claim is made about corruption types not tested. On these types ($n = 300$ each), each LG hypothesis was rejected with $b > c$. Under shot noise accuracy rose from 0.673 to 0.767 (net 0.093, 0.047 to 0.143; one-sided $p = 0.0002$, Holm-adjusted $p = 0.0005$, two-sided $p = 0.0003$), and under speckle noise from 0.697 to 0.757 (net 0.060, 0.017 to 0.103; one-sided $p = 0.006$, Holm-adjusted $p = 0.011$, two-sided $p = 0.011$). Under impulse noise it rose from 0.700 to 0.750 (net 0.050), and the 95% interval has lower bound 0.0 (0.000 to 0.100). The one-sided and Holm-adjusted p-values are both 0.029, and the two-sided p is 0.058, so this cell would not pass a two-sided 0.05 test, and no two-sided claim is made for it. The three cells are reported separately and are not pooled.

TABLE II
BASELINE MODELS: REFERENCE ACCURACY OF CLIP VIT-L/14 AND SIGLIP SO400M PER CONDITION

Condition	n	CLIP acc.	CLIP drop (95% interval)	SigLIP acc.	SigLIP drop (95% interval)
Clean images	450	0.998	–	0.998	–
Gaussian noise, $\sigma = 0.1$	450	1.000	−0.002 [−0.007, 0.000]	0.998	0.000 [0.000, 0.000]
JPEG, quality 10	450	0.998	0.000 [0.000, 0.000]	1.000	−0.002 [−0.007, 0.000]
Splice, 128 px	450	0.996	0.002 [0.000, 0.007]	1.000	−0.002 [−0.007, 0.000]
Clean images, same 300 items	300	0.997	–	0.997	–
Shot noise, severity 3	300	1.000	−0.003 [−0.010, 0.000]	0.997	0.000 [0.000, 0.000]
Impulse noise, severity 3	300	0.997	0.000 [0.000, 0.000]	0.997	0.000 [0.000, 0.000]
Speckle noise, severity 3	300	1.000	−0.003 [−0.010, 0.000]	0.997	0.000 [0.000, 0.000]
Cost per item	– not measured (no timing in any committed baseline run)				
Pixel parity, 5 items, the four conditions of the top block: minimum token agreement 0.989 (threshold 0.98)					
Pixel parity, 5 items, clean and three severity-3 conditions: minimum token agreement 0.986 (threshold 0.98)					

For reference only: different models trained on different data; no test and no comparison are made. Same first 450 fold items (300 for the severity-3 cells) and regenerated pixels; revisions and scores in supplement S-IV. Drop: clean minus corrupted accuracy of the same model on the same items, paired percentile bootstrap 95% interval. The row "Clean images, same 300 items" was computed at writing from the stored predictions (descriptive). Cost per item was not measured.

E. Reference Encoders

For reference, CLIP ViT-L/14 [65] and SigLIP so400m [66], different models trained on different data, reached accuracies of 0.996 to 1.000 on the same items in every condition (Table II), with the same decision rule on images regenerated with the runner's corruption functions and seeds (supplement S-IV). Pixel parity was shown: the UniDisc tokeniser's re-encoding of the regenerated images agreed with the stored codes on at least 98.6% of tokens (five items per condition set; threshold 98%). Nearly every cell is at or next to its ceiling, so no reading beyond the stated accuracies is drawn, and no test is made.

F. Cost

The fresh-fold pass, which computes the unweighted rule for four conditions and stores the inputs of LRF, used 67 network rows and a median of 8.0 s per item on one 12 GB laptop GPU, with a peak memory of 8.2 GB; the stage of the other noise types used 42 rows and 5.0 s per item (supplement S-II). Per scored image, the unweighted sum needs two caption-visible passes with the whole image masked, so that their outputs do not depend on the image codes (the evidence of every condition is a gather at its codes). LRF needs those two and nine more for its features: eight complementary-mask passes (for B and the uncertainty of the B pass) and one observed-position pass; D comes from a cached caption-masked null of fixed slot length, and the texture and codebook-neighbour features need no pass. LRF therefore costs $2 + 9 = 11$ passes per scored image against 2, and then computes its weights and decision from the stored arrays on the CPU: the pinned scorer took 6.5 s for all 1 800 image-and-condition decisions of the fresh

fold (3.6 ms each, start-up included) on the authors' laptop. The count follows the number of mask groups ($K = 8$), not the $m = 1\,024$ grid positions; another K or grid size was not tried.

G. Exploratory Ablation

Exploratory; no ablation was preregistered, and supplement S-V plots and tabulates every variant. The following intervals contained zero: those of a linear logit on the same 35 features, of the features computed from B and D alone, of the features from the uncertainty of the B pass alone, and of attention-weighted pooling in the style of gated-attention multiple-instance learning (MIL) [47]. They lay below zero for the other single-source weightings, the unlearned rank weight, and a single-feature weight (the distance from the grid centre, chosen from 35 candidate features by the inner-fold objective on the design items, a selection budget 35 times that of the frozen LRF; the selection used no fresh item). Among knockouts, the intervals for the features from B and D and for the neighbourhood features lay below zero and the interval for the codebook-neighbour-distance features above it; every other group's interval contained zero. Setting a group to its mean gives inputs that the model did not see together in fitting; an interval away from zero shows that the frozen model's decisions depend on those inputs, not that the group alone detects anything. Across the three refit seeds, the range of fresh-fold accuracy was at most 0.030 in every variant and cell; seed 0 was fixed before any fresh item and is not the best of three. The 336 intervals are dependent and unadjusted.

H. Comparison With Alternative Aggregation Rules (Exploratory)

Exploratory; no test, and the marks in Table III are descriptive. Table III sets the accuracy of the unweighted sum and of LRF beside other ways of combining the same per-token evidence Δ_j on the 450 fresh-fold images: three rules without a fitted weight (a rank weight, the median, and the 20% trimmed mean, defined in the table note) and four weightings fitted on the 500 design items (one feature, a linear logit on the 35 features, an additive model on two feature groups, and attention-weighted pooling [47]). The median, the trimmed mean, and a Dempster combination of evidence-derived token opinions had lower accuracy than no weighting in every condition. The linear logit reached 0.884 under JPEG against 0.864 for LRF (difference 0.020, unadjusted 95% interval 0.004 to 0.038) and 0.931 under the splice against 0.927, and the additive model on two feature groups reached 0.922 on clean images against 0.918. Denoising before scoring (non-local means; the last two rows of Table III) gave, under Gaussian noise, 0.844 with the unweighted sum and 0.856 with LRF, which is 0.042 (unadjusted 95% interval 0.002 to 0.082) and 0.053 (0.011 to 0.096) above LRF without denoising (0.802); under JPEG quality 10 it gave 0.853 and 0.813, which are -0.011 (-0.044 to 0.024) and -0.051 (-0.089 to -0.013) from LRF. Nothing here favours the additive form over a linear logit on the same features; the confirmed object is the frozen weighting, and the preregistered result remains the L1 test of Table I. The additive form is kept because each feature's learned shape can be read directly (Fig. 5); this is a reason of interpretability, not of accuracy.

TABLE III
EXPLORATORY COMPARISON WITH BASELINE METHODS THAT COMBINE THE SAME PER-TOKEN EVIDENCE: ACCURACY ON THE 450 FRESH-FOLD IMAGES

Method	Noise 0.1	JPEG 10	Splice 128	Clean
Unweighted sum†	0.733	0.849	0.896	0.887
Unlearned rank weight	0.731	0.838	0.891	0.896
Median [67]	0.716	0.829	0.864	0.862
20% trimmed mean [67]	0.716	0.827	0.873	0.869
Dempster combination [37]	0.722	0.838	0.889	0.884
Single feature (selected)	0.742	0.840	0.891	0.893
Linear logit, 35 features	0.791	*0.884*	*0.931*	0.920
Additive, B and D only [64]	0.800	0.871	0.911	*0.922*
Attention pooling (MIL) [47]	0.793	0.838	0.851	0.864
LRF (frozen)†	0.802	0.864	0.927	0.918
Denoise [68], unweighted sum	0.844	0.853	–	–
Denoise [68], LRF	*0.856*	0.813	–	–

†Fixed before the fold was opened (only the unweighted sum and the frozen LRF); every other row was specified and scored after the preregistered results. Italics: highest accuracy in the column (ties share the mark); descriptive marks of one sample, not a test. 4 of the 9 learned refits of the exploratory ablation are shown, chosen after the fold was scored; all 10 are in supplement S-V, with differences and intervals. Median: $\mathrm{med}_j\,\Delta_j > 0$; 20% trimmed mean: the mean of Δ_j without its $\lfloor 0.2m \rfloor$ lowest and highest values is positive ($m = 1\,024$). Unlearned rank weight: $w_j = 1 - r_j$, $r_j \in [0, 1]$ the within-image percentile rank of the token's uncertainty (one minus the largest predicted probability, caption masked). Dempster combination: per token, evidence $\max(\pm\Delta_j, 0)$ with Dirichlet prior weight 2 gives an opinion, discounted by w_j of the rank weight and combined over tokens by Dempster's rule; the opinions are constructed from Δ_j, not learned by an evidential head. Denoise: non-local-means denoising before scoring (setting fixed in advance), noise and JPEG only. LRF was not trained on denoised images; the denoise-then-LRF row applies the frozen weights outside their training distribution.

LRF was chosen by inner cross-validation among four learned families on the design items (Section IV-A), and the learned shapes in supplement S-II are descriptive. In an exploratory sweep of the Gaussian noise level on the same 450 images (one noise draw per image, scaled by σ; Fig. 6 and supplement S-V), the frozen LRF exceeded the unweighted sum at $\sigma = 0.05$, 0.2, and 0.3 by 0.056 (unadjusted 95% interval 0.020 to 0.091), 0.058 (0.018 to 0.098), and 0.084 (0.044 to 0.124); no level showed a loss. On clean images the difference was 0.031 (0.004 to 0.058), and the excess of the gain under noise over the clean gain excluded zero only at $\sigma = 0.3$ (0.053, 0.009 to 0.098; at the fitted level 0.038, -0.004 to 0.080), so the part of the gain that is specific to noise is clear only at the strongest level.

V. DISCUSSION

A. Reading the Result

L1 shows that one frozen weighting, hash-pinned before any test image was chosen, raised image-direction verification accuracy under Gaussian noise with $\sigma = 0.1$ on 450 COCO images that played no part in its design or fit, passing an exact paired test at the preregistered level and holding with length-matched negative captions. It does not show that the weighting works on a corruption it was not trained on, that its weights follow the reliability of a token, or that the effect carries to another model, dataset, or severity. Six inexpensive alternative

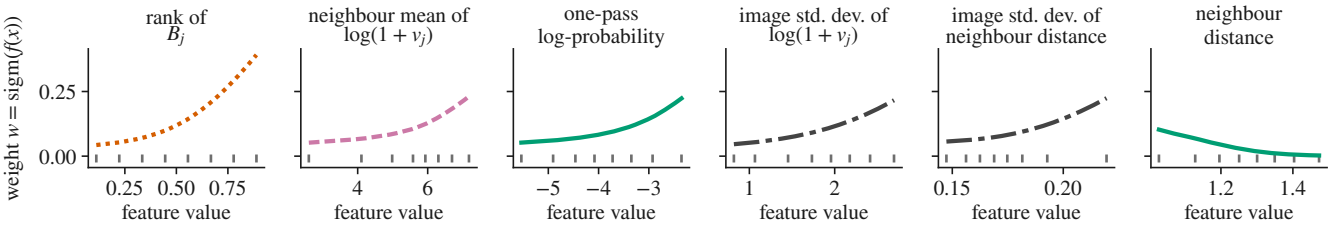


Fig. 5. Learned shapes of the frozen weight model: partial dependence of the weight w on the six of 35 features with the largest learned effect range, every other standardised feature at its mean 0 (knots: grey ticks; line style: feature group of Fig. 2). Descriptive; not evidence that the weights track corruption.

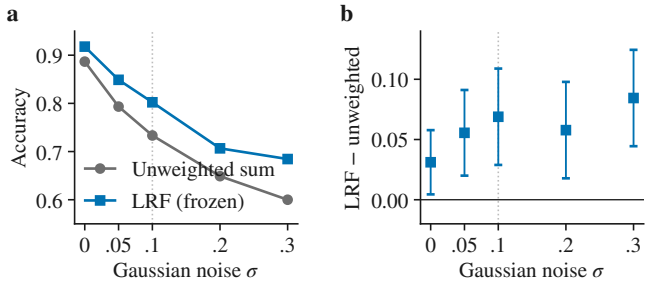


Fig. 6. Exploratory Gaussian-noise severity sweep on the 450 fresh-fold images (one noise draw per image, scaled by σ): (a) accuracy of the unweighted sum and the frozen LRF; (b) LRF minus the unweighted sum with its unadjusted 95% paired percentile-bootstrap interval. Dotted line: the confirmatory level $\sigma = 0.1$, the only level LRF was trained on; levels 0 and 0.1 are the confirmatory run's cells. No test.

readings remain open, each with its evidence in Section V-B: (1) a reweighting that helps whatever the condition; (2) a supervised fit to the training corruption; (3) a single noise-sensitive feature, such as the codebook-neighbour distance or the one-pass observed-position score, which the learned weight shapes (supplement S-II) cannot separate and on which the ablation of Section IV-G bears only as an exploratory result; (4) a limited reach to other noise types, since the impulse-noise interval starts at 0.0; (5) an unstable splice decision, whose Holm-adjusted p of 0.049 lies 0.001 below the level; and (6) a gain under Gaussian noise that a generic pixel-domain denoiser applied before scoring exceeded in this sample, with an exploratory, unadjusted interval excluding zero (Section IV-H), so that this paper does not show learned weighting to be the preferable defence against this noise. A second backbone or a corruption held out of training would settle these questions; neither was run.

B. Limitations

One model, one dataset, and one severity per corruption were tested in the preregistered tests; the exploratory severity sweep adds three Gaussian levels on the same images, not new images. LRF was trained on the tested corruption (Gaussian noise with $\sigma = 0.1$ was a training condition), so L1 tests new images, not a new corruption. No preregistered test separates corruption-specific weighting from a generic gain, since the clean-image and splice estimates were also positive; "under Gaussian noise" names the tested condition, not a specificity. The Holm-adjusted p of L4 is 0.049, and the two-sided p of the impulse-noise cell is 0.058. The clean-image margin of L-NI was fixed after an exploratory clean interval had been seen. Whether COCO images were in the pretraining data of UniDisc, its tokeniser, CLIP, or SigLIP is unknown. Each image has one corruption realisation and one mask seed, so the intervals cover item sampling only. A fresh tokenisation of the same pixels in another process need not reproduce the stored codes: it agreed on 98.8% and 98.9% of image tokens in the two stopped self-checks (supplement S-I, deviation 1), and on at least 98.6% in the pixel-parity runs. The reference encoders almost solve this matching task (Section IV-E); this paper makes no claim of a verifier that ranks with them. No second backbone was scored, since a likelihood score of a quantised adapter would be confounded and LRF would need refitting; no second dataset and no hard-negative captions were tested; and the scorer is a sum of one-pass conditionals, a factorised surrogate, so an exact ordered likelihood was not computed (Section III-B). No ablation and no comparison with other aggregation rules was preregistered; Sections IV-G and IV-H are exploratory. In that ablation, simpler weightings fitted on the design items (a linear logit, the features from B and D alone, the features from the uncertainty of the B pass alone, and attention-weighted pooling) were not distinguishable from the frozen LRF within their intervals under Gaussian noise. The linear logit on the same 35 features was at or above the frozen LRF in three of the four conditions (accuracies in Table III) and above the unweighted rule with a 95% interval excluding zero in all four, including JPEG (difference 0.036, 0.007 to 0.064; unadjusted, one of 40 exploratory variant-by-condition cells), where the preregistered test of LRF detected no difference. A non-local-means denoiser applied before scoring reached higher accuracy than LRF under Gaussian noise (Section IV-H; exploratory, unadjusted), so the paper does not show that LRF is the better choice against this noise.

All results are for category-disjoint negatives (a random caption of an image sharing no annotated COCO category with the test image). The preregistration is self-hosted (Section III-E), and the audits were run automatically in the authors' own pipeline (not independent reviews).

C. Deviations From the Plans

(1) The L1 stage stopped twice at its tokenisation self-check before any item; a dated amendment moved the code pin, and a second changed one test only. (2) The scorer's pin file was written while

L1 ran, with every value copied from records committed before L1. (3) The environment lock files of the design, re-score, L1, and other-noise stages are empty; the environment was recorded after the runs, and the corruption package's commit was verified from it. (4) Item arrays are untracked with committed SHA-256 manifests; hashes appear in CRLF and LF forms of the same content. (5) The pixel-parity files were moved unchanged from a wrong output directory with no prior hash (audit-time hashes recorded); they supersede the main reference-encoder file's field "pixel_parity: not run". (6) The reference-encoder download was 5.2 GB, against about 2 to 4 GB quoted to the first author. (7) What had been seen before the re-freeze is stated in Section IV-A. (8) "No output was read before its run was complete" rests on the operations log. (9) The exploratory ablation, the exploratory comparison with alternative aggregation rules (Section IV-H, Table III; plan committed before any value was computed), the qualitative figure (Fig. 4; selection rule fixed after the confirmatory result), the Dempster-combination row, and the severity sweep (both with plans committed before any value was computed) were added at writing, as exploratory additions. The plans for the denoise rows, the Dempster-combination row and the severity sweep were committed before their runs; the environment lock files of the denoise and sweep runs are empty, and the environment record predates them. (10) 2026-10-31: this paper reports one of two preregistered studies; the other is not reported here. The analysis addendum (A6.12) had placed a sentence on the other study in this paper's abstract, so this is a deviation from it; the methods state that study's negative primary result (A6.13) without its numbers. (11) 2026-10-31: the title was shortened from the title preregistered for the confirmed branch; the claim is unchanged. (12) 2026-10-31: the abstract contains no numbers; they are in the Results. This departs from the sentence templates of A6.12, which carried the numbers in the abstract; no claim was added. Supplement S-I gives each item in full with its record.

VI. Conclusion

We tested one frozen, learned reliability weighting of the per-token evidence of a pretrained unified masked diffusion model on a fresh preregistered fold. Under Gaussian noise with $\sigma = 0.1$ it raised image-direction verification accuracy from 0.733 without reliability weighting to 0.802 (net 0.069, 95% interval 0.029 to 0.109; exact one-sided McNemar $p = 0.0005$), including with length-matched negatives, and on clean images it lost no more than three percentage points (0.03 in accuracy; 90% interval of the net change 0.009 to 0.053); the margin was fixed after an exploratory clean interval had been seen. This holds for one model, on 450 COCO images not used to design or fit the method, at one severity; the weighting was trained on the tested corruption, and no preregistered test shows that its gain is specific to that corruption.

Reproducibility and Data Availability

The code, the preregistration documents (cards, analysis plans, and their addenda), the hash pins, the recorded predictions, and the analysis outputs are kept in one version-controlled repository with its commit history (github.com/shlokDS16/reliability-aware-fusion-unidisc); the commit identifiers printed in this paper and in supplement S-I refer to it. A release of the repository is archived on Zenodo (doi: 10.5281/zenodo.23810711). The per-item arrays are not tracked; their SHA-256 manifests are committed. The software environment was recorded after the runs. The images and captions are the public COCO val2017 split [60], and the model is the released UniDisc checkpoint [4] at the snapshot pinned in supplement S-I.

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